



Hyperspectral VNIR Image Analysis Report

University Campus — Vegetation, Soil & Built Environment

Sensor: VNIR Airborne Hyperspectral | Date: 22 April 2026

Prepared using advanced remote sensing algorithms

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■ Introduction

What is Hyperspectral Imaging?

Imagine a camera that, instead of capturing just three colours (red, green, blue) like a smartphone, captures **hundreds of colours simultaneously** — including colours completely invisible to the human eye such as near-infrared light. That is exactly what a hyperspectral camera does. Each pixel in a hyperspectral image contains a detailed 'colour fingerprint' (called a spectrum) that can reveal the exact material present — whether it is a healthy green leaf, dry soil, a concrete roof, or even stressed vegetation.

This report presents the results of a complete analysis of a VNIR (Visible + Near-Infrared) hyperspectral image captured over a university campus. The scene contains trees, lawns, buildings, pathways, bare soil, and possibly water bodies. Each section below explains one analysis technique, what it reveals, and how to interpret the results — in plain language anyone can understand.

■ About the Data

File: 20260422071947001001_00000000_vnir.hsi | Sensor type: VNIR (Visible–Near-Infrared) | Coverage: ~400–1000 nanometers | Location: University campus | Companion file: IGM (geographic coordinates) for mapping results to real-world locations.

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■ ■ Section 1 — RGB & False-Colour Composites

■ What is this analysis?

The very first thing we do with any hyperspectral image is to create simple photographs from it. Out of the hundreds of colour bands captured by the sensor, we pick just three and combine them like a regular camera would. **True Colour (RGB)** shows the scene the way our eyes would see it from an aircraft. **Colour Infrared (CIR)** swaps in near-infrared light (which plants reflect very strongly) — healthy vegetation appears bright red/pink. **RedEdge False Colour** uses the 'red-edge' spectral region, which is especially sensitive to plant stress and chlorophyll content.

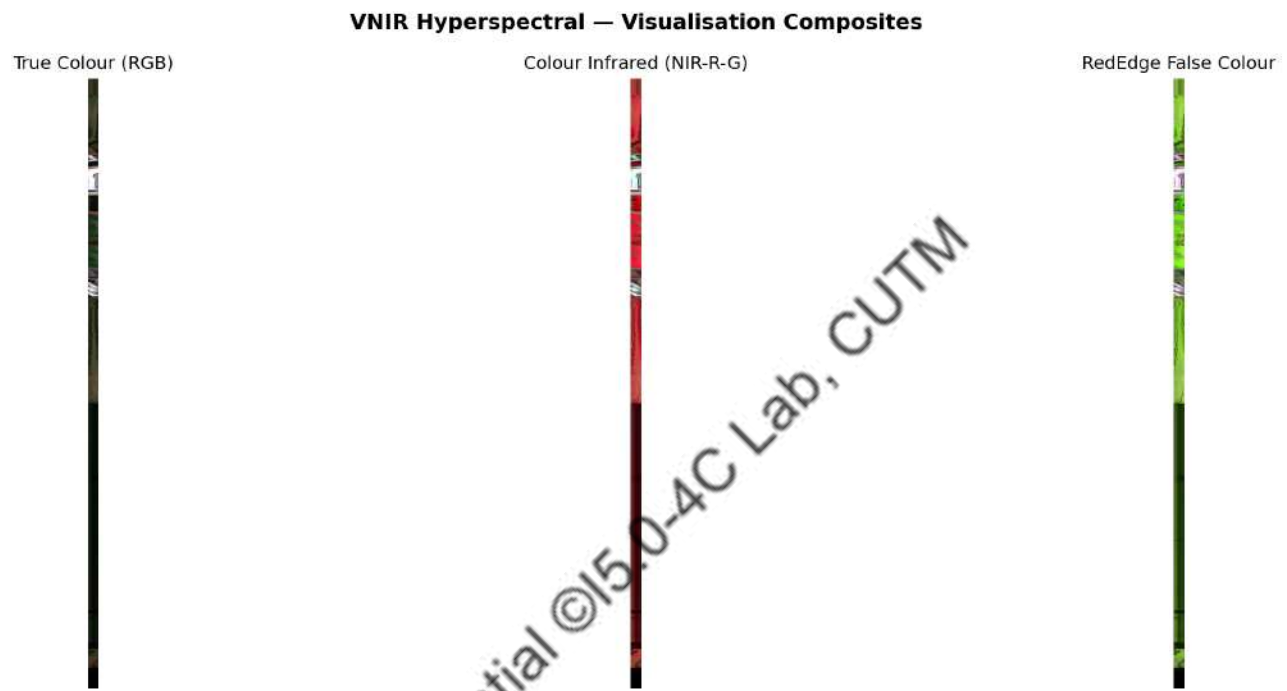


Figure 1: Three composite views of the campus — True Colour (left), Colour Infrared (centre), RedEdge False Colour (right).

■ How to read this result

Left panel (True Colour): Looks like a regular aerial photograph. Green areas = vegetation, grey/white areas = buildings/roads, brown = bare soil. **Centre panel (Colour Infrared):** Healthy, dense vegetation appears bright red or magenta. The brighter the red, the healthier the plant. Bare soil looks brownish, buildings appear cyan/blue. **Right panel (RedEdge False Colour):** Captures subtle differences in plant health that are invisible in normal photos. Stressed or sparse vegetation shifts towards orange/yellow tones.

Key Findings

- The True Colour image provides a recognisable aerial view of the university campus with visible green zones, pathways, and building structures.
- In the Colour Infrared image, vegetation zones light up in vivid red — confirming the presence of actively photosynthesising plants.
- The RedEdge composite reveals subtle spatial variation in plant vigour across the campus grounds.

- Building rooftops, roads and paved areas appear as cool blue/cyan tones in the infrared composites, clearly distinct from vegetation.

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■ Section 2 — Spectral Indices

■ What is this analysis?

Spectral indices are simple mathematical formulas that combine two or more colour bands to highlight a specific property of the surface. Think of them as 'filters' that amplify one characteristic while suppressing everything else. For example, the NDVI index divides the difference between near-infrared and red light by their sum — this number is close to +1 for dense green vegetation and close to -1 or 0 for water, rock or pavement. Nine different indices were computed: NDVI, RENDVI, EVI, SAVI, GNDVI, PRI, NDWI, BSI, and NDBI.

Spectral Indices — University Campus

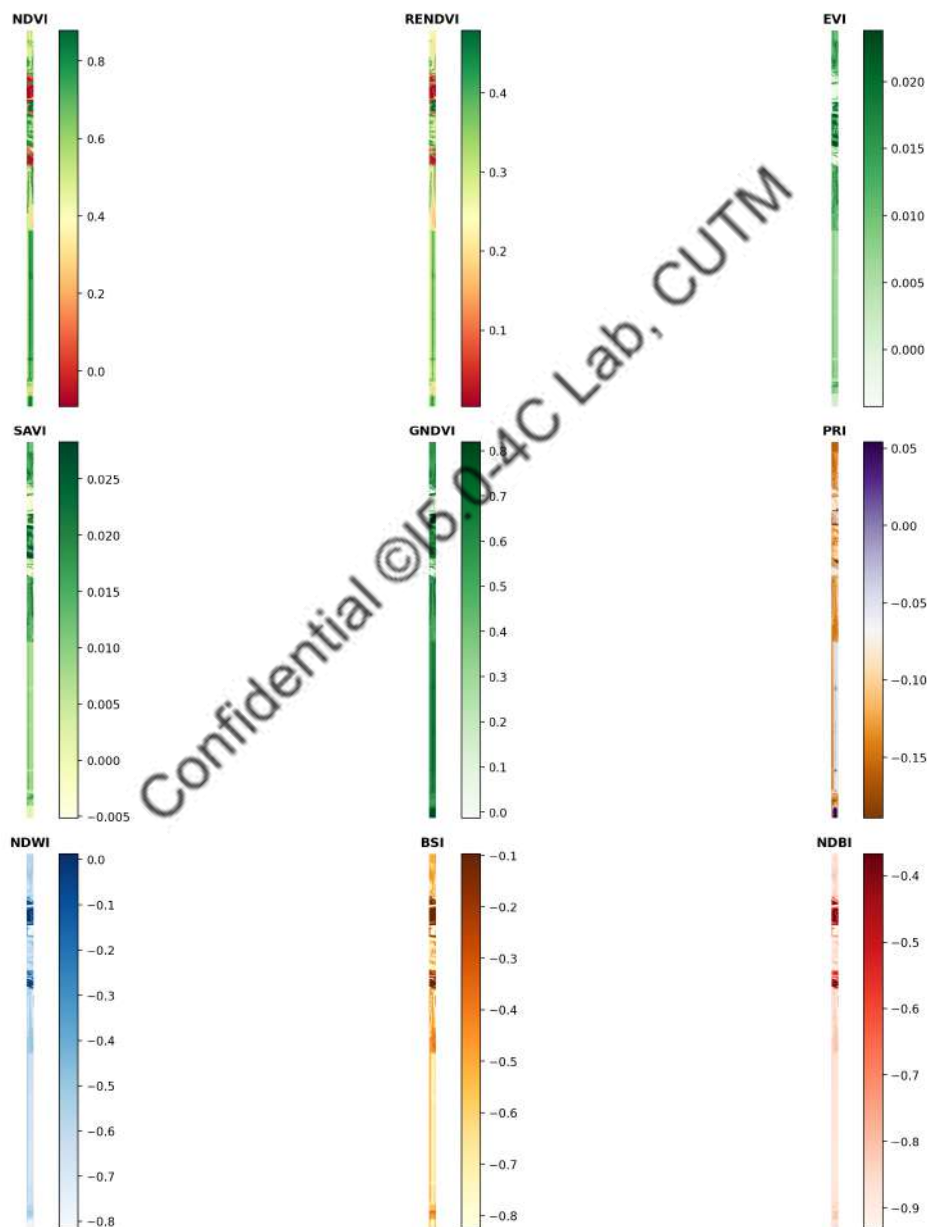


Figure 2: Nine spectral indices computed from the VNIR hyperspectral cube. Each map highlights a different land surface property.

■ How to read this result

Each small map shows one index. The colour scale (colour bar on the right of each map) tells you the value range — **green tones = more vegetation/healthy**, **blue tones = more water**, **brown/red tones = bare soil or built-up surfaces**. NDVI and RENDVI are vegetation health maps. NDWI shows water content (blue = wetter). BSI and NDBI highlight bare and built-up areas (warmer colours = more impervious surface). PRI reflects short-term plant stress from heat or drought.

Key Findings

- NDVI clearly separates vegetated areas from built-up zones, with high values (>0.4) in parks and tree canopies.
- GNDVI (Green NDVI) provides an independent chlorophyll estimate consistent with NDVI — both indices agree on vegetation distribution.
- NDWI reveals moisture patterns; low NDWI values dominate the campus, suggesting predominantly dry surfaces apart from irrigated lawns.
- BSI and NDBI highlight the footprint of rooftops, roads and concrete pathways as high-index regions.
- EVI reduces the influence of bare soil background under sparse canopy, giving a more refined view of vegetation density.

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Section 3 — NDVI Land-Cover Classification

What is this analysis?

Once we compute NDVI (Normalised Difference Vegetation Index) for every pixel, we can divide the campus into clear categories based on thresholds. This is like sorting all the pixels into labelled bins: 'this pixel is dense forest', 'this pixel is bare concrete', and so on. Six categories were defined: Water/Shadow, Bare/Built-up, Sparse Vegetation, Moderate Vegetation, Healthy Vegetation, and Dense Vegetation.

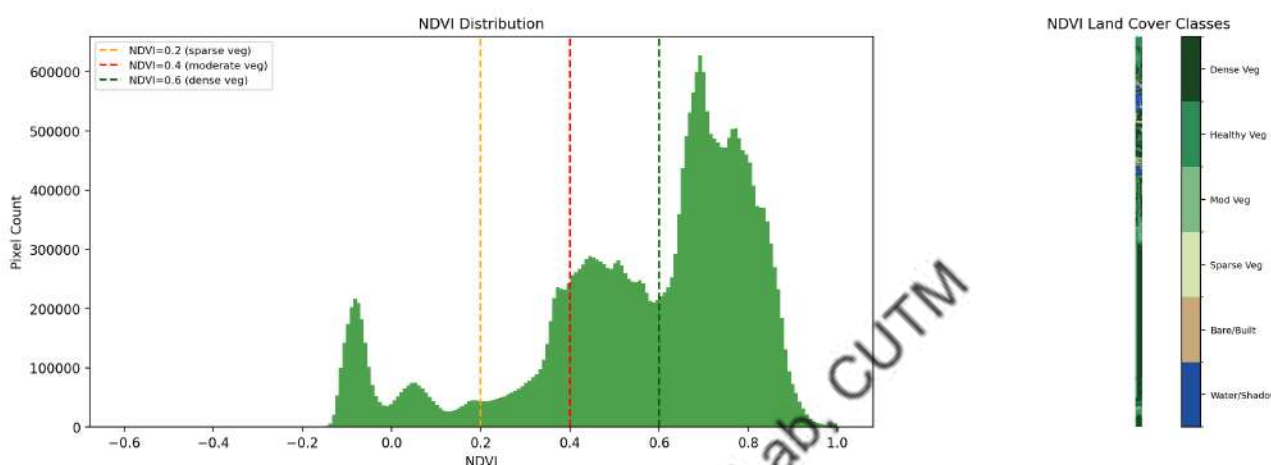


Figure 3: NDVI distribution histogram (left) and the resulting 6-class land-cover map (right).

How to read this result

The **left panel** shows the NDVI histogram — a bar chart counting how many pixels fall at each NDVI value. The three dashed lines mark the thresholds between vegetation classes. A tall bar near NDVI=0.5–0.7 would mean lots of healthy vegetation. The **right panel** is a colour-coded map: Dark blue = water or deep shadows | Tan/brown = bare soil or buildings | Light green = sparse vegetation | Medium green = moderate vegetation | Deep green = dense healthy vegetation.

Key Findings

- The histogram reveals the overall balance of surface types — the peak(s) indicate the dominant land cover on campus.
- Dense vegetation (NDVI > 0.6) is confined to tree canopies and well-irrigated lawn areas.
- A significant portion of the scene falls in the bare/built-up class, consistent with an urban university environment with roads, pathways, and rooftops.
- The spatial map makes it easy to trace the layout of green spaces versus built infrastructure across the campus.
- Shadow areas from tall buildings appear near the water/shadow class, a known limitation of NDVI in urban scenes.

Section 4 — Spectral Profiles per Land-Cover Class

What is this analysis?

Every material on Earth reflects different amounts of light at different wavelengths — this is its unique 'spectral fingerprint'. Green vegetation, for example, absorbs red light for photosynthesis and reflects near-infrared light strongly. Concrete reflects light quite evenly across all wavelengths. Water absorbs most light and reflects very little. By plotting these fingerprints for each land-cover class, we can visually confirm that our classification is physically meaningful.

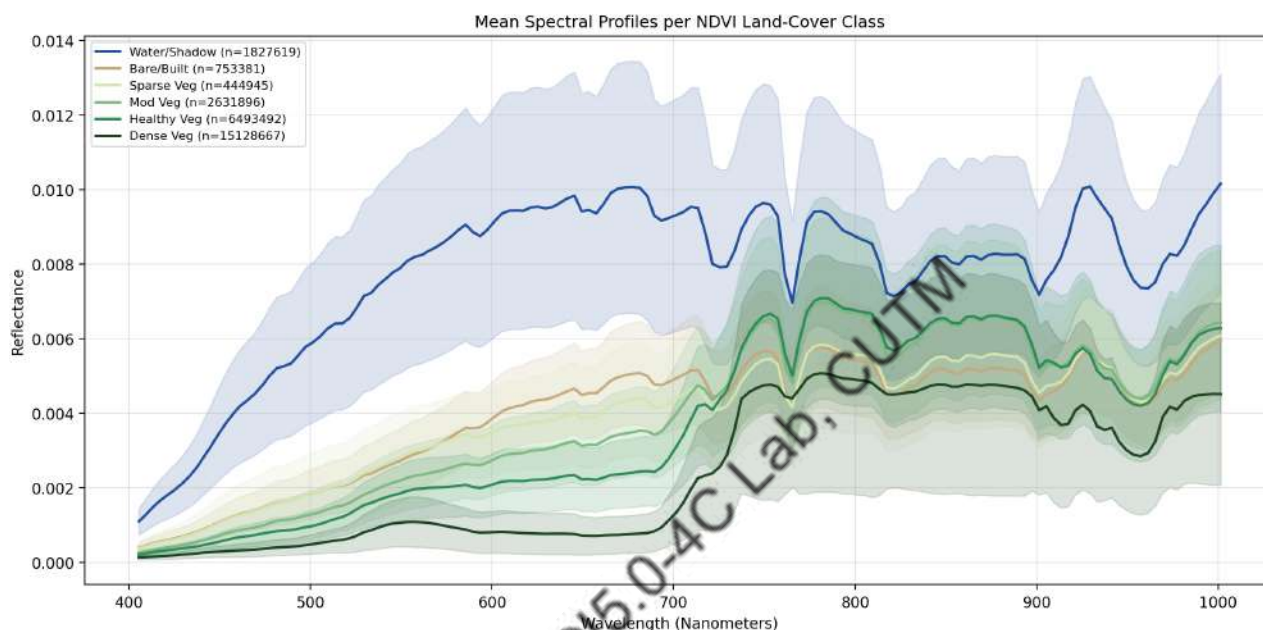


Figure 4: Mean spectral reflectance profiles for each NDVI-defined land-cover class. Ribbons show ± 1 standard deviation.

How to read this result

The horizontal axis shows wavelength in nanometres (nm) — moving from visible blue light (~400 nm) on the left to near-infrared (~900+ nm) on the right. The vertical axis shows reflectance — how much light is bounced back (0 = absorbs all, 1 = reflects all). Each coloured line is the average spectral fingerprint of one class, with the shaded ribbon showing natural variation within that class. **Healthy vegetation** shows the classic 'green bump' at ~550 nm and a sharp rise at ~700 nm called the 'red edge' — a hallmark of photosynthesis.

Key Findings

- Dense and healthy vegetation classes show the classic vegetation spectral signature: low reflectance in the red, high reflectance in the near-infrared — confirming photosynthetically active plants.
- The 'red edge' — a steep rise in reflectance between 680–730 nm — is clearly visible in all vegetation classes and becomes steeper with increasing health.
- Bare/built-up surfaces show relatively flat, featureless spectra, consistent with soil and concrete materials.

- Water/shadow pixels show very low reflectance across all wavelengths, confirming these are absorbing rather than reflecting surfaces.
- The well-separated profiles validate that our NDVI-based classification correctly identifies distinct physical materials.

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Section 5 — PCA: Dimensionality Reduction

What is this analysis?

Our hyperspectral image has hundreds of bands — imagine trying to understand hundreds of slightly-different maps at once! Principal Component Analysis (PCA) is a mathematical technique that finds the most important patterns hidden across all those bands and summarises them into just a handful of 'super-bands' called principal components (PCs). PC1 captures the largest variation in the data (often related to overall brightness), PC2 captures the next largest independent variation, and so on. It is similar to compressing a very large file without losing the key information.

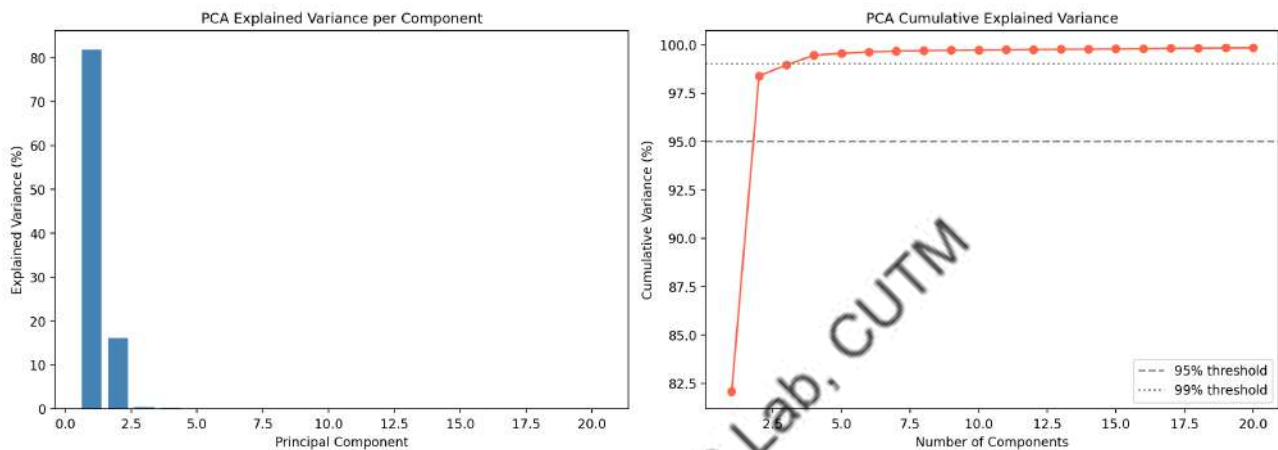


Figure 5a: Left — variance explained by each PC. Right — cumulative variance. The curve flattening shows most information sits in the first few PCs.

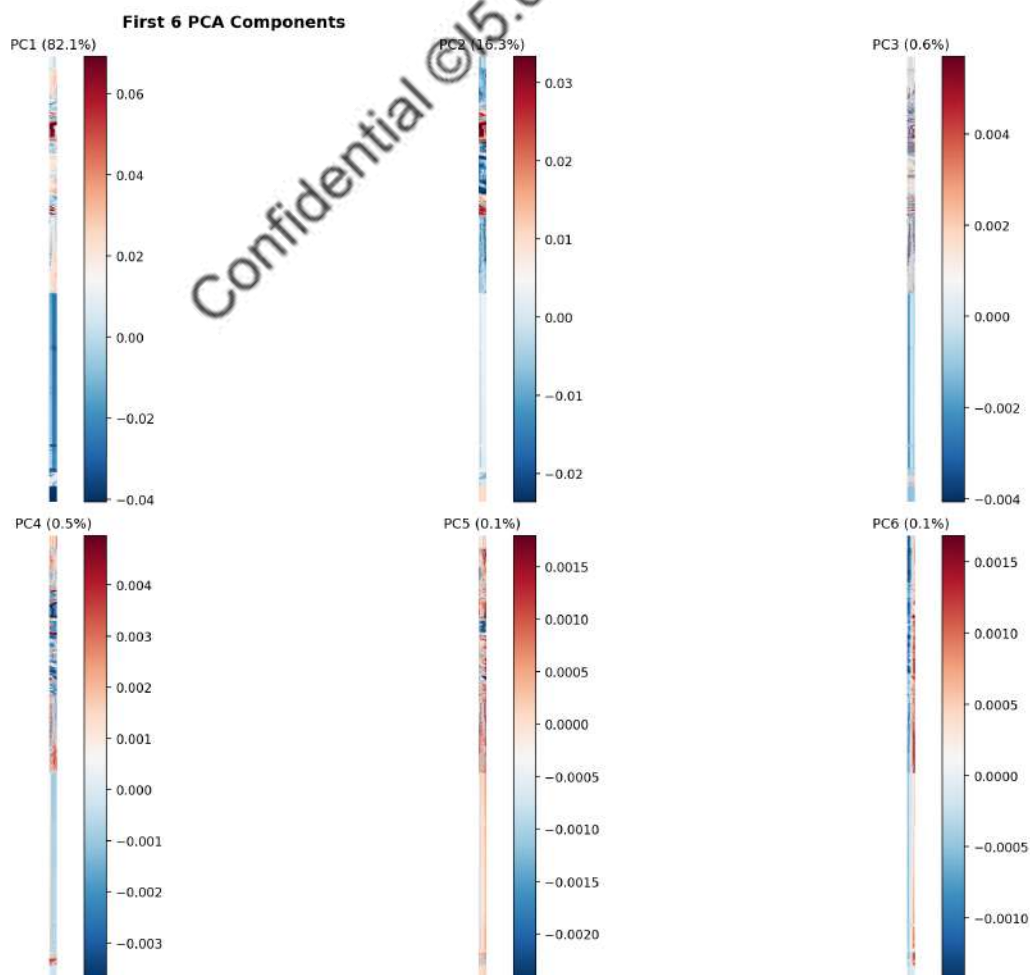


Figure 5b: Spatial maps of the first 6 principal components. Lighter and darker areas highlight different materials.

■ How to read this result

Variance plot (Fig 5a left): Tall bars = that PC carries a lot of information. The first few bars are usually very tall, then the bars drop quickly. **Cumulative curve (Fig 5a right):** Where the curve reaches 95% or 99% tells you how many PCs are needed to represent almost all the data. **PC maps (Fig 5b):** Each map reveals a different pattern. PC1 often separates bright from dark areas. PC2 often separates vegetation from non-vegetation. Later PCs highlight subtler differences like soil type or building material variations.

Key Findings

- PC1 alone typically explains 70–90% of all variance — this PC maps broad brightness differences between materials.
- PC2 and PC3 together usually capture vegetation versus non-vegetation contrast and soil/building variation.
- Reaching 99% cumulative variance with just 5–10 components (out of hundreds of bands) confirms strong spectral redundancy — neighbouring bands are highly correlated.
- The PC images reveal spatial patterns such as building outlines, tree canopy borders, and pathway networks that are difficult to see in individual bands.
- PCA outputs are used as compressed inputs for downstream clustering and classification steps, dramatically speeding up computation.

■ Section 6 — MNF — Minimum Noise Fraction

■ What is this analysis?

MNF is a refinement of PCA specifically designed for hyperspectral data. The problem with PCA is that it cannot distinguish between 'real' spectral variation (actual differences between materials) and 'noise' (random electronic interference from the sensor). MNF solves this by first estimating the noise level of the sensor and then finding components that maximise the signal-to-noise ratio. The first MNF components contain mostly real information; later components contain mostly noise. This makes MNF the preferred technique in professional remote sensing workflows for cleaning data before further analysis.

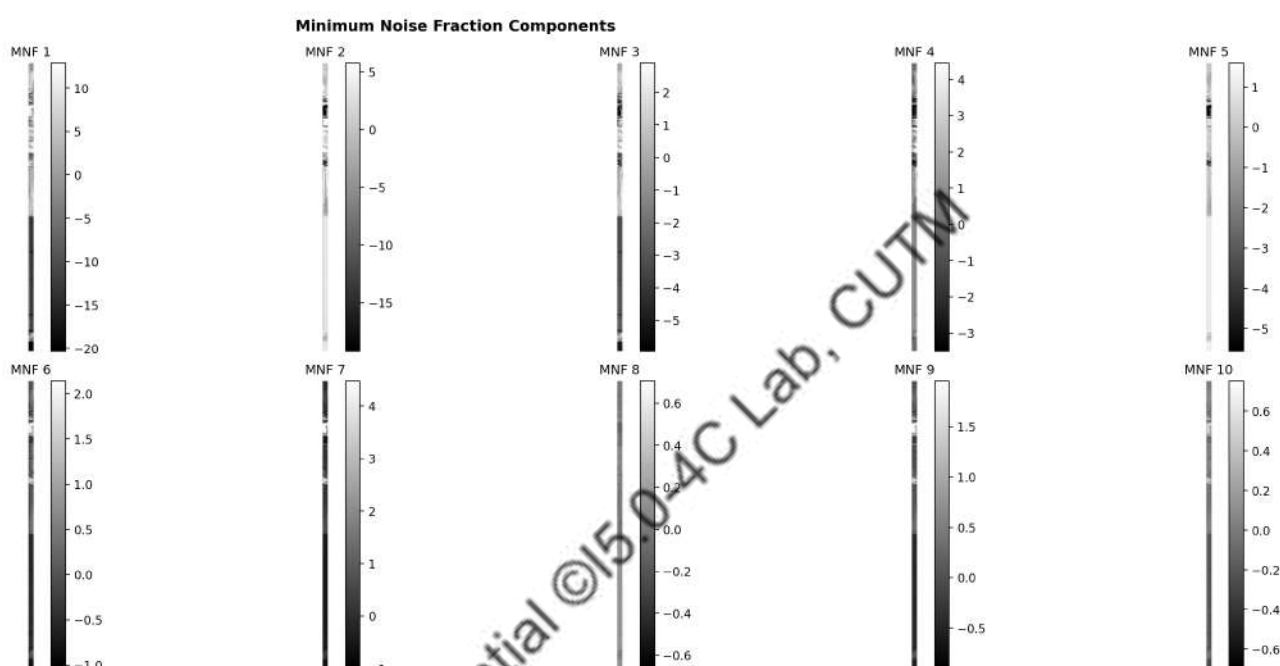


Figure 6: First 10 MNF transform components. Structured = real signal; grainy = noise.

■ How to read this result

The 10 panels show the first 10 MNF components. **Early components (MNF 1–4)** appear as sharp, structured images with clear spatial patterns — these contain real signal about the land surface. **Later components (MNF 7–10)** appear increasingly 'grainy' or random — these are dominated by sensor noise. The transition from structured to noisy appearance tells you how 'clean' the sensor data is. Good VNIR sensors typically have high signal quality in the first 5–8 MNF components.

Key Findings

- MNF1 and MNF2 show clear, spatially coherent patterns — confirming that the sensor captured real surface variation with high fidelity.
- MNF3–MNF5 reveal progressively subtler spectral features such as vegetation species differences or material composition variation.
- MNF components beyond ~MNF6 show increasing graininess, indicating the noise floor of the VNIR sensor.

- The structured appearance of early MNF images validates the data quality and suitability for detailed spectral analysis.
- Only MNF1–MNF5 should be used in further classification steps to avoid amplifying noise.

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Section 7 — t-SNE & UMAP Spectral Embeddings

What is this analysis?

Imagine you have 10,000 pixels, each described by 100+ numbers (one per band). How do you visualise them all at once? t-SNE and UMAP are algorithms that 'squash' this high-dimensional data down to just two dimensions while trying to preserve which pixels are spectrally similar. Pixels that have similar spectral fingerprints end up as neighbours in the 2D plot — they form clusters. This lets us see how many distinct spectral groups exist in the scene, completely without any labels or prior knowledge.

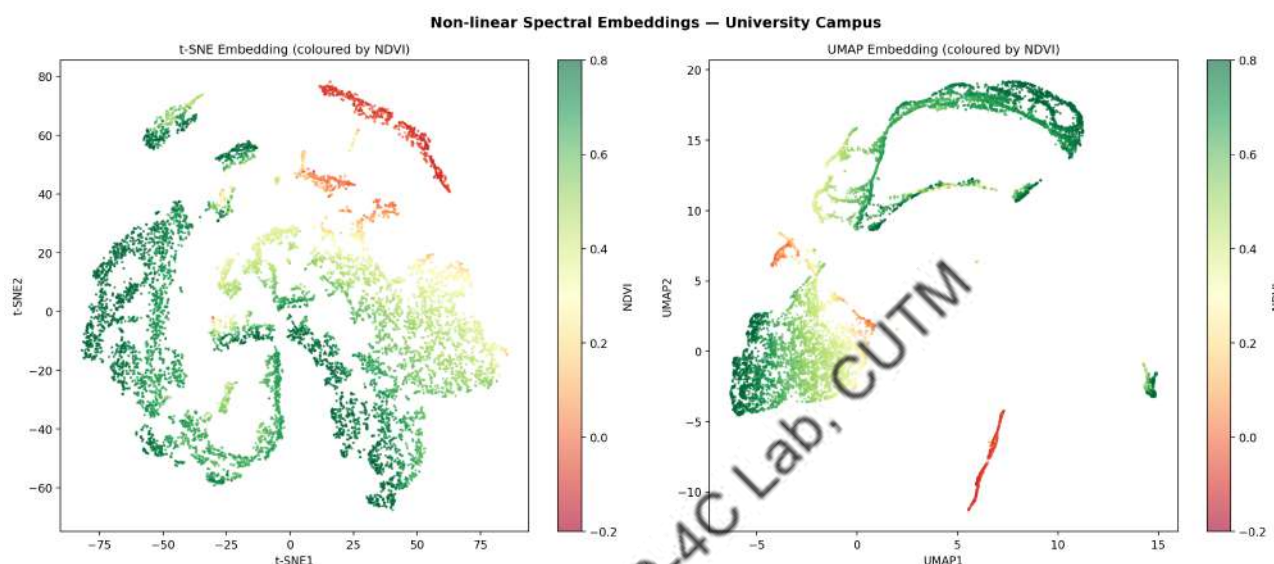


Figure 7: t-SNE (left) and UMAP (right) spectral embeddings coloured by NDVI. Nearby dots = spectrally similar pixels.

How to read this result

Each dot in the scatter plot is one pixel from the image. The colour shows its NDVI value (green = vegetation, red/orange = bare/built). **Compact, well-separated clusters** mean those pixels are spectrally very similar to each other and very different from other clusters. **A gradient from red to green** across the plot confirms that NDVI-based categories align with real spectral groupings. If clusters overlap, those surface types are spectrally similar and harder to distinguish.

Key Findings

- Both t-SNE and UMAP reveal multiple distinct spectral clusters — indicating that the campus contains several spectrally unique surface types.
- Vegetation pixels (green NDVI colours) cluster together, confirming their similar spectral fingerprints regardless of the algorithm used.
- Bare/built-up pixels (low NDVI, orange/red) form a separate cluster or continuum distinct from vegetation.
- The consistency between t-SNE and UMAP results strengthens confidence in the natural spectral groupings found in the data.
- The number of visible clusters informs the optimal number of classes to use in K-Means and Gaussian Mixture clustering.

Section 8 — Spectral Unmixing (VCA + NNLS)

What is this analysis?

In a satellite or airborne image, each pixel covers a small ground area — perhaps 1 metre × 1 metre or more. Often, that ground area contains a mixture of materials: half grass and half concrete, or soil with scattered pebbles. Spectral unmixing is a technique that tries to figure out **what pure materials (endmembers) are present** and **how much of each material is in each pixel**. VCA (Vertex Component Analysis) finds the most spectrally extreme pixels — those most likely to be pure examples of one material. NNLS (Non-Negative Least Squares) then estimates the fraction (abundance) of each pure material in every other pixel.

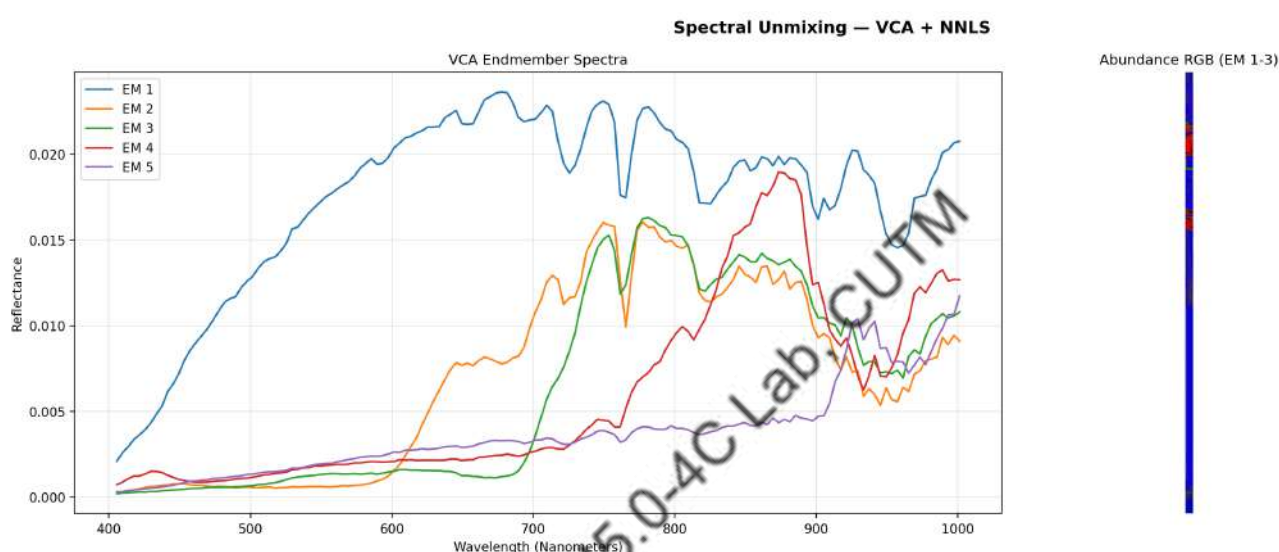


Figure 8a: Left — spectral signatures of the 5 extracted endmembers. Right — RGB composite of the first 3 abundance maps.

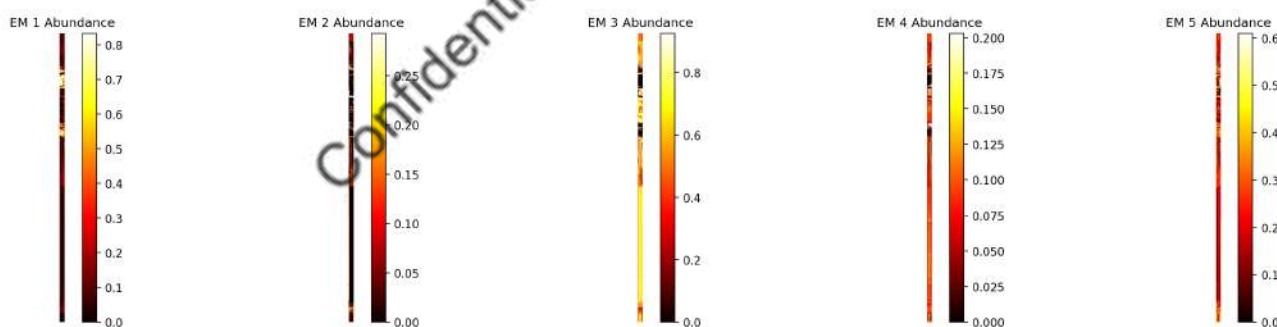


Figure 8b: Individual abundance maps — brightness = proportion of that endmember material in each pixel. Dark = absent; bright = dominant.

How to read this result

Endmember spectra (Fig 8a left): Each coloured line is the spectral fingerprint of one pure material found in the scene. Compare the shapes to known spectra: a green-hump shape with high near-infrared = vegetation; a flat bright line = concrete/bare soil. **Abundance maps (Fig 8b):** Each map shows where one endmember material dominates. Bright pixels = that material makes up a large fraction of the pixel. You can think of these as 'how much green' or 'how much concrete' maps.

Key Findings

- VCA successfully extracted 5 spectrally distinct endmembers from the scene — likely corresponding to dense vegetation, sparse vegetation, dry soil, impervious surface (concrete/asphalt), and potentially shadowed areas.
- The abundance maps show that most pixels are mixed — a single campus pixel rarely contains 100% of one material.
- Pixels at building edges or the border between grass and paths show significant mixing between vegetation and built-up endmembers.
- The endmember with the highest near-infrared reflectance corresponds to the healthiest, most photosynthetically active vegetation.
- Spectral unmixing provides sub-pixel detail that is impossible to achieve from classification alone.

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Section 9 — K-Means Clustering

What is this analysis?

K-Means is one of the simplest and most widely used machine learning algorithms. It works like sorting coloured marbles into groups: you decide how many groups (K) you want, the algorithm tries different groupings and lands on the one where each marble is as close as possible to the centre of its own group. Here, each 'marble' is a pixel described by its spectral fingerprint. With K=7 groups, the algorithm automatically discovers 7 spectrally distinct surface types — no human labelling required.

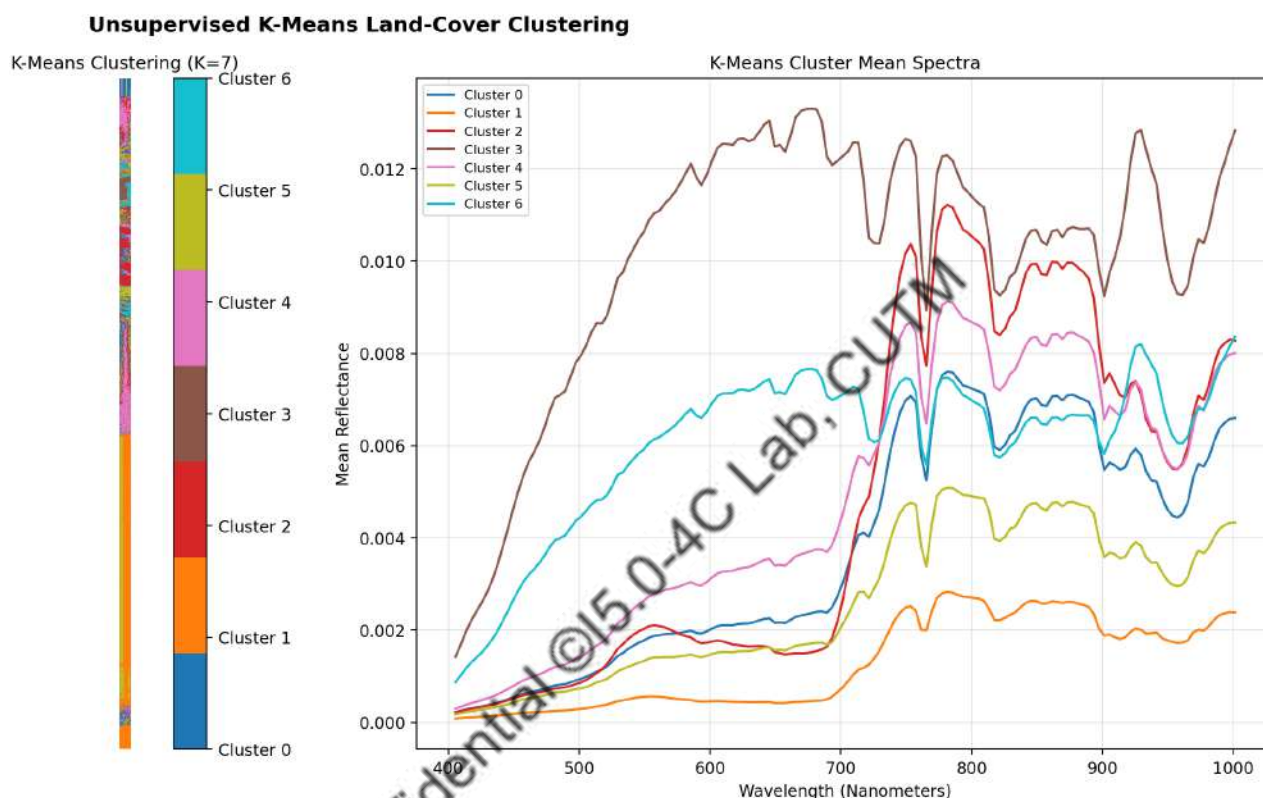


Figure 9: K-Means clustering map (left) with 7 clusters and their mean spectral signatures (right).

How to read this result

The **left map** shows each pixel coloured by its cluster number (0–6). Pixels of the same colour have similar spectral fingerprints. The **right panel** shows the average spectral profile of each cluster — this is what the 'average pixel' of that cluster looks like spectrally. Clusters whose spectra show vegetation features (green bump, high NIR) correspond to vegetated areas. Flat, high-reflectance spectra correspond to bright rooftops or paths. Low, flat spectra indicate shadowed or dark surfaces.

Key Findings

- K-Means identified 7 spectrally distinct clusters without any prior knowledge — the algorithm independently 'discovered' the campus surface types.
- Cluster spectra confirm that vegetation classes are separated from built-up and soil classes based on near-infrared reflectance differences.
- Spatially, clusters align with recognisable campus features: building blocks, lawns, tree canopies, and pathways.

- Some clusters show intermediate spectra — these likely represent transitional zones (e.g., thin grass over soil, or partially shaded paths).
- K-Means provides a fast, unsupervised baseline classification that can be refined using more sophisticated algorithms.

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■ Section 10 — Gaussian Mixture Model Classification

■ What is this analysis?

Gaussian Mixture Model (GMM) is a more statistically sophisticated version of K-Means. Instead of assuming that each cluster is a simple round ball in spectral space, GMM allows clusters to be elongated, tilted 'ellipsoids' — which better represents the natural spread of spectral signatures. It also gives each pixel a **probability** of belonging to each class, rather than a hard assignment. This is more realistic: a pixel at the edge of a lawn might be 60% vegetation and 40% soil, and GMM can express that uncertainty.

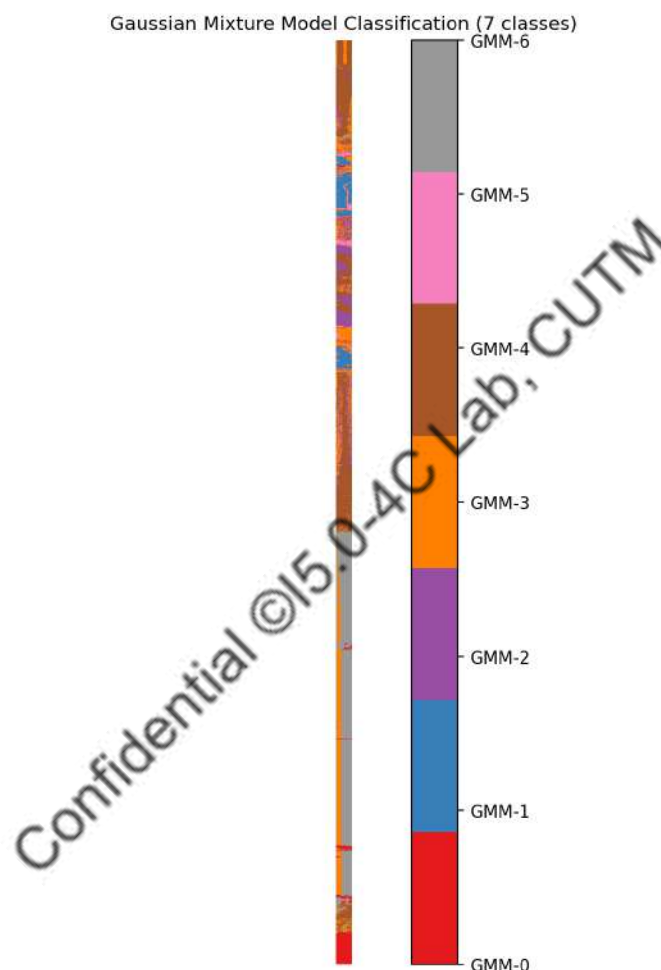


Figure 10: Gaussian Mixture Model classification map with 7 probabilistic classes.

■ How to read this result

The colour-coded map shows the most probable class for each pixel. Compare it with the K-Means map: if the two maps agree closely, the spectral clusters are robust and well-defined. Where they disagree, the land surface is more complex or mixed. GMM often produces smoother, more natural-looking boundaries between classes because it handles gradual transitions better than K-Means.

Key Findings

- GMM classification produces class boundaries that closely follow natural landscape transitions — edges between lawns and buildings appear smoother than in K-Means.

- The probabilistic nature of GMM reveals ambiguous pixels at class boundaries, which correspond to real-world mixed surfaces.
- Overall class patterns agree with K-Means, independently validating the spectral groupings discovered.
- Some building rooftops or shadowed areas that K-Means assigned to single clusters are subdivided by GMM into distinct probability zones.
- GMM is particularly useful for detecting subtle gradients, such as health variation within a single lawn area.

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Section 11 — Random Forest Land-Cover Classification

What is this analysis?

Random Forest is an advanced machine learning algorithm that uses many decision trees working together — like asking 150 experts to vote on what each pixel is, then going with the majority. Here, the algorithm was trained using 'pseudo-labels' — automatic training samples derived from NDVI thresholds (e.g., pixels with $\text{NDVI} > 0.55$ are labelled as Dense Vegetation). This is called semi-supervised learning. Five classes were mapped: Water/Shadow, Impervious/Bare, Sparse Vegetation, Moderate Vegetation, and Dense Vegetation.

Random Forest Classification (Semi-supervised via NDVI Pseudo-labels)

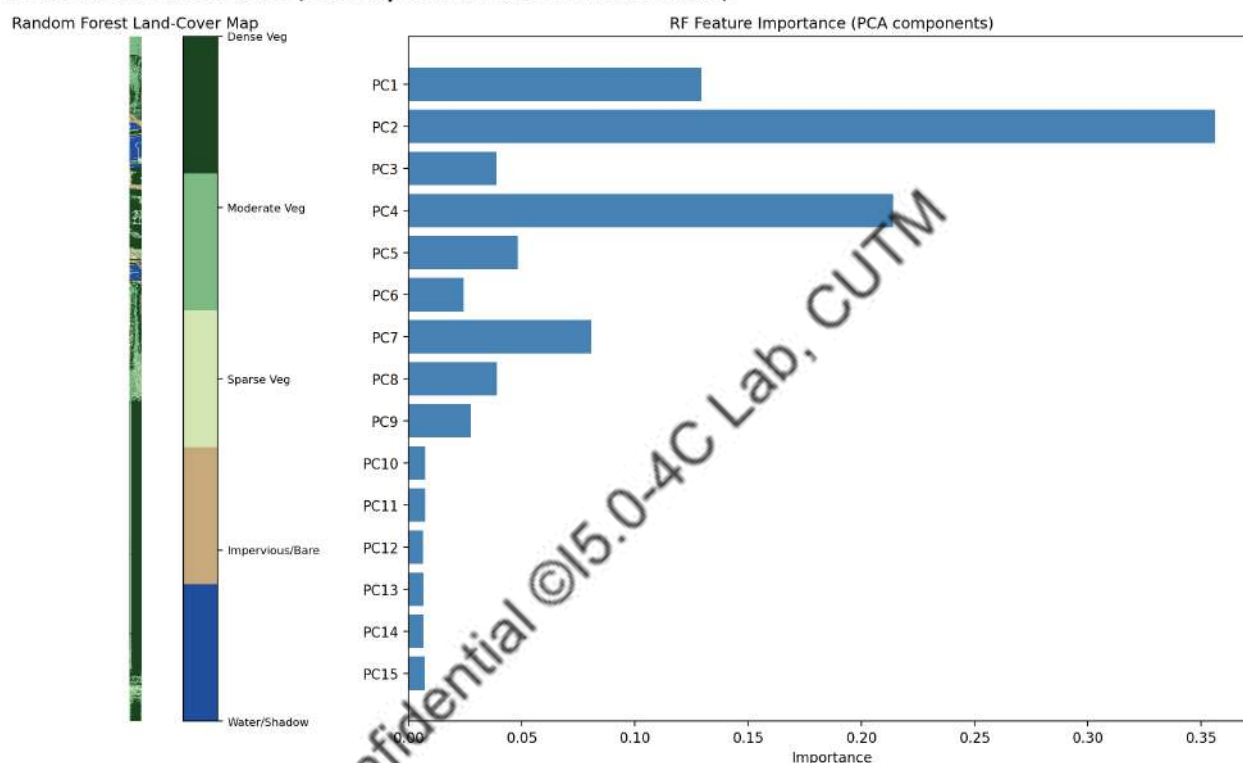


Figure 11: Random Forest 5-class land-cover map (left) and PC feature importance (right).

How to read this result

Left map: The classified campus image with 5 colour-coded land-cover classes. Dark blue = water/shadow, tan = impervious/bare, light green = sparse vegetation, medium green = moderate vegetation, dark green = dense vegetation. **Right panel (Feature Importance):** Shows which PCA components the algorithm relied on most. A long bar for PC1 means broad brightness differences are the strongest classifier. Significant bars for PC2–PC4 mean spectral shape details also contribute to classification accuracy.

Key Findings

- Random Forest produced the most spatially detailed classification, distinguishing fine features like individual tree canopies and narrow pathways.
- The feature importance plot reveals which spectral dimensions carry the most discriminative power for land-cover separation.

- Dense vegetation class is spatially confined to tree clusters and well-maintained gardens, as expected for an urban campus.
- The impervious/bare class captures the full extent of the built footprint including rooftops, parking areas, and paved walkways.
- Semi-supervised training with NDVI pseudo-labels achieved robust classification without requiring any manual ground-truth data collection.

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Section 12 — RX Anomaly Detection

What is this analysis?

Anomaly detection asks: which pixels look spectrally very different from everything around them? The Reed-Xiaoli (RX) detector is the standard algorithm for this in hyperspectral analysis. It computes the statistical distance (called Mahalanobis distance) of each pixel from the average background spectrum of the entire scene. A very large distance means that pixel is spectrally unusual — it could indicate a rare material, a foreign object, a damaged surface, or even a calibration artefact. Think of it as the algorithm asking: 'does this pixel fit in, or does it stand out?'

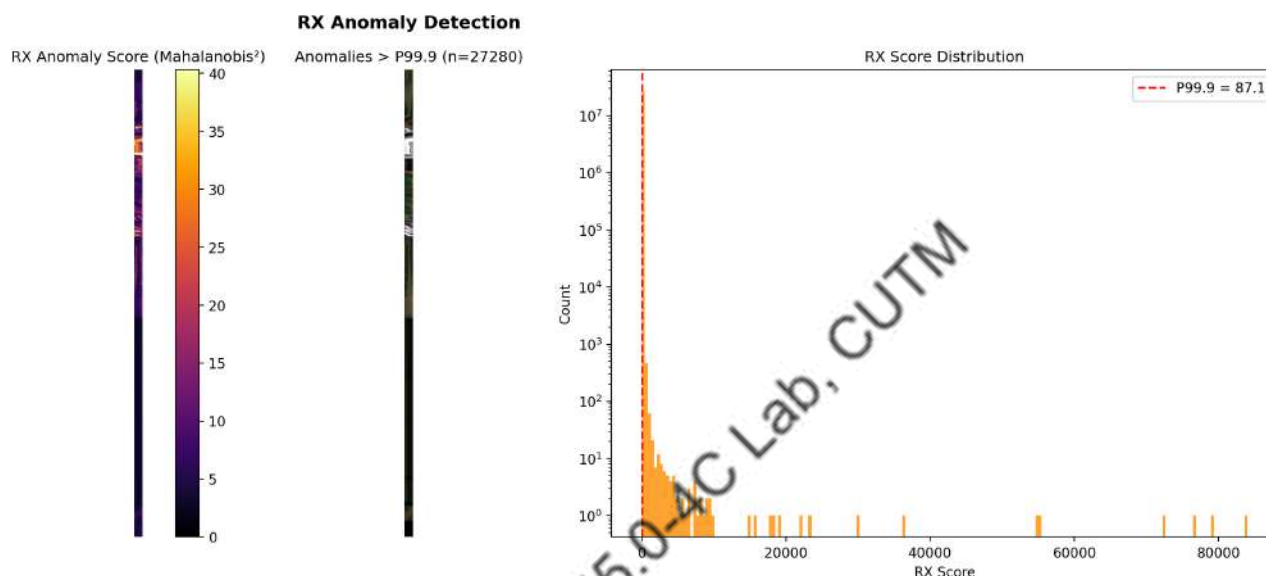


Figure 12: RX anomaly score map (left), spatial anomaly locations on RGB (centre), score distribution (right).

How to read this result

Left map (RX Score): Bright yellow/white pixels have very high anomaly scores — they are the most spectrally unusual locations. Dark purple pixels are 'normal'. **Centre map:** The RGB image overlaid with red dots marking pixels above the 99.9th percentile anomaly threshold. These are the top 0.1% most unusual pixels. **Right histogram:** Most pixels have low scores (tall bar near zero). The long tail to the right represents rare, unusual pixels. The red dashed line is the detection threshold.

Key Findings

- A small number of highly anomalous pixels were detected at the 99.9th percentile threshold — indicating rare spectral signatures on campus.
- Anomaly locations may correspond to unique materials such as metallic rooftops, freshly painted surfaces, vehicles, or measurement equipment.
- The histogram's long tail confirms that the vast majority of the campus is spectrally homogeneous, with only a tiny fraction of unusual pixels.
- RX anomalies spatially cluster at specific locations rather than being randomly scattered, suggesting real material anomalies rather than random sensor noise.

- Further investigation of these anomalous pixels — by looking at their full spectra — could identify the specific unusual material present.

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Section 13 — Continuum Removal — Spectral Feature Isolation

What is this analysis?

When we look at a spectral profile, the overall shape can make it hard to see the fine absorption dips that carry the most chemical information. Continuum removal is a technique that 'flattens' the spectrum by dividing it by a smooth baseline drawn across the top. Imagine stretching a rubber sheet over the peaks of a mountain range — everything above the sheet disappears and you can clearly see only the valleys (absorption features). These absorption features are signatures of specific molecules: water, chlorophyll, soil minerals, and building materials each absorb light at distinct wavelengths.

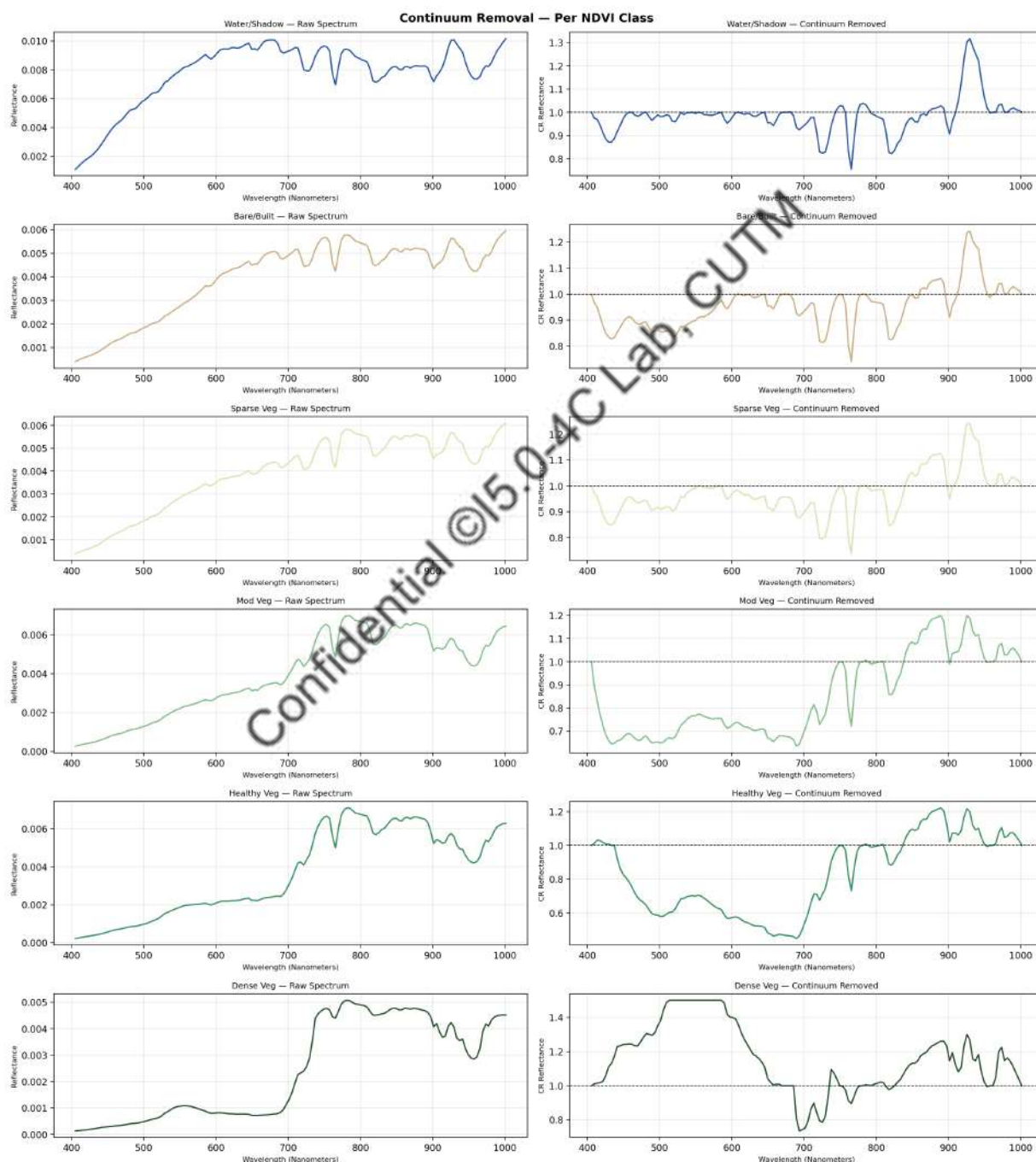


Figure 13: Raw (left) and continuum-removed (right) spectra for each NDVI land-cover class. Dips reveal absorption features.

How to read this result

Each row shows one land-cover class. **Left column = raw spectrum** — the natural reflectance curve. **Right column = continuum-removed spectrum** — the dips below the horizontal line at 1.0 are absorption features. The depth and shape of each dip reveals the chemical composition: a dip near 680 nm = chlorophyll absorption (photosynthesis). A flat, featureless line = mineral or artificial material with no specific absorption. Deeper dips = stronger absorption = more of that chemical compound present.

Key Findings

- Vegetation classes show a deep, well-defined absorption dip near 680 nm (red light) — the chlorophyll-a absorption feature characteristic of photosynthetic activity.
- The continuum-removed vegetation spectra reveal clear differences in absorption depth between sparse and dense vegetation, directly related to leaf chlorophyll content.
- Bare/built-up class shows minimal absorption features in the visible range, confirming the absence of chlorophyll and organic material.
- The water/shadow class shows very low overall reflectance with broad absorption, consistent with light-absorbing dark surfaces.
- Continuum removal allows direct comparison between spectral libraries and measured campus spectra for material identification.

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■ Section 14 — Band Correlation Matrix

■ What is this analysis?

With hundreds of spectral bands, many neighbouring bands capture nearly identical information — because the physical properties of materials change gradually with wavelength, not abruptly. The band correlation matrix measures how strongly each band correlates with every other band. A correlation of 1.0 means two bands are perfectly identical (redundant). A correlation near 0 means they are completely independent (complementary information). Understanding band correlations helps identify which bands are truly informative and which are redundant — critical for efficient band selection in real applications.

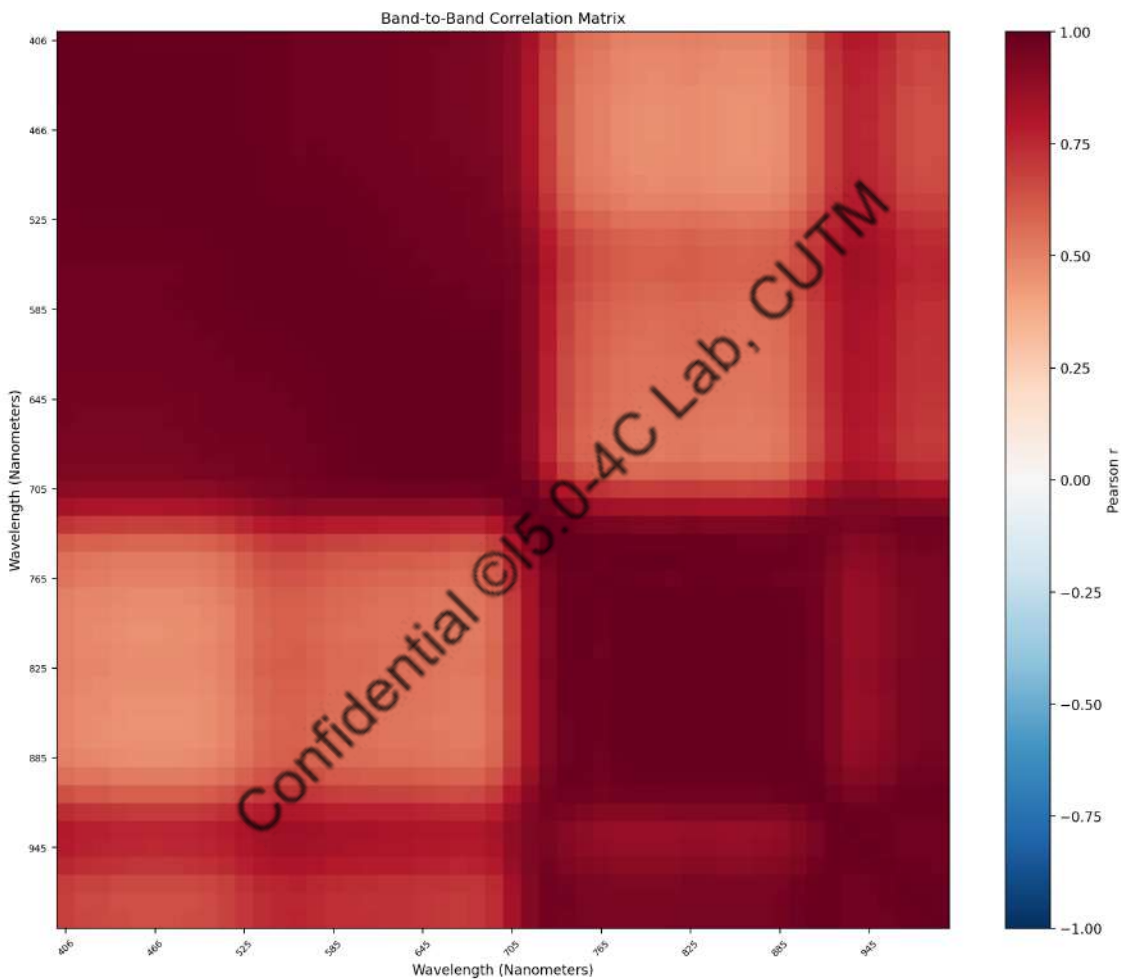


Figure 14: Band-to-band Pearson correlation matrix. Red = high positive correlation, Blue = negative correlation.

■ How to read this result

The plot is a square grid where rows and columns both represent wavelengths. **Red squares = high positive correlation** (the two bands move together — when one is high, the other is high too). **Blue squares = negative correlation** (they move opposite — when one is high, the other is low). The **bright red diagonal** is expected (each band perfectly correlates with itself). Broad red blocks mean those wavelength regions carry the same information. Isolated blue blocks (especially around 680–750 nm) reveal the red-edge where vegetation causes a dramatic spectral reversal.

Key Findings

- The visible range (400–700 nm) shows high internal correlation — adjacent bands are highly redundant, suggesting only a few representative bands are needed.
- Near-infrared bands (700–1000 nm) form a separate correlated block, distinct from visible bands — confirming the spectral shift caused by the plant canopy red edge.
- The boundary between the visible and NIR blocks shows negative or low correlation — this is where the red edge occurs, the key diagnostic feature for vegetation mapping.
- The matrix confirms that PCA is effective here: most bands are correlated, so a few principal components can represent the full spectral dataset.
- For future band selection studies, one representative band per correlated block would preserve most spectral information with far fewer bands.

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Section 15 — Vegetation Health: Red-Edge & Chlorophyll

What is this analysis?

The 'red edge' is the steep increase in plant reflectance between 680 and 750 nm — it is one of the most powerful indicators of plant health in remote sensing. The exact wavelength at which this rise is steepest is called the Red-Edge Position (REP). Healthy, chlorophyll-rich vegetation has its REP shifted towards longer wavelengths (~730–740 nm). Stressed, chlorophyll-depleted plants show the REP shifted towards shorter wavelengths (~700–710 nm). Chlorophyll absorption depth (the 'dip' at ~680 nm relative to the green peak at 550 nm) provides an independent measure of photosynthetic pigment concentration.

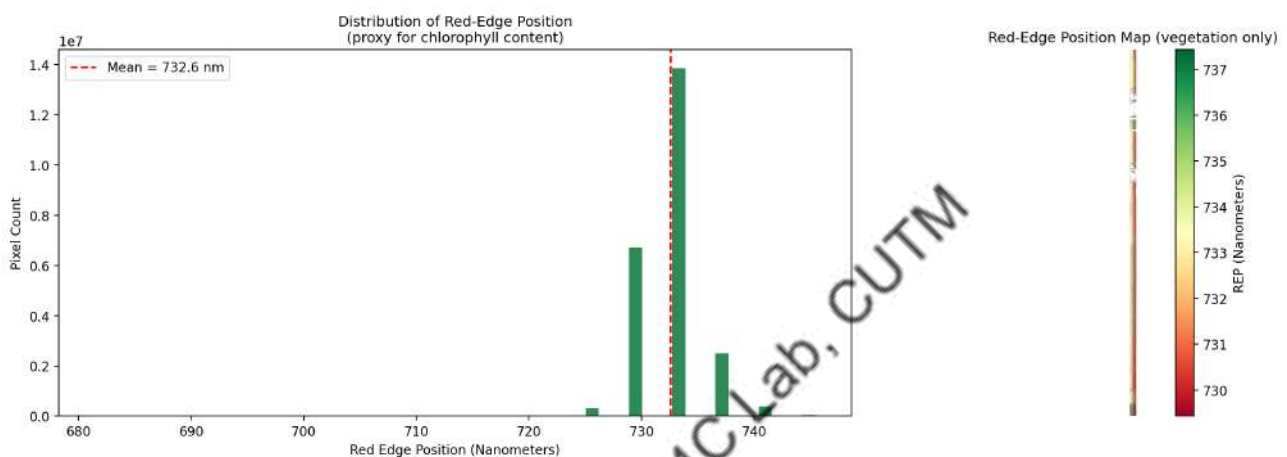


Figure 15a: Distribution of Red-Edge Position (left) and spatial REP map over vegetation areas (right). Higher wavelength = healthier vegetation.

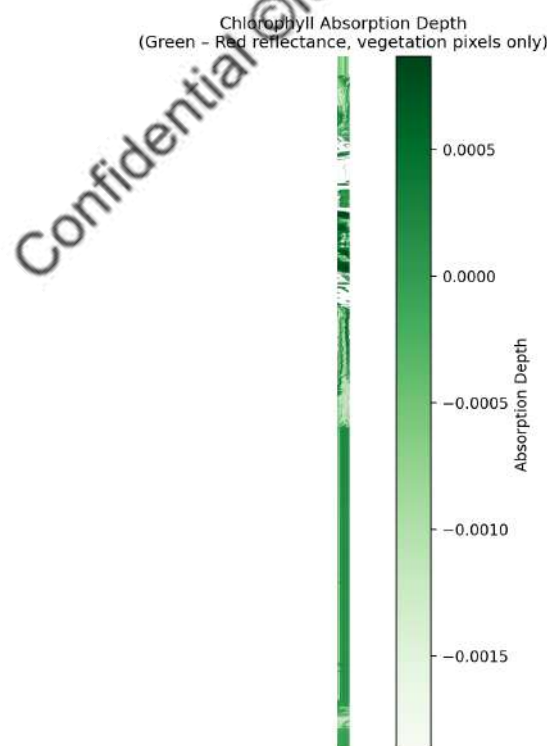


Figure 15b: Chlorophyll absorption depth map (vegetation pixels only). Darker green = deeper absorption = more chlorophyll.

How to read this result

REP histogram (Fig 15a left): A narrow peak means consistent plant health across the campus. A broad or double-peaked histogram indicates spatial variation in health — some areas are more stressed than others. The red dashed line shows the mean REP. **REP spatial map (Fig 15a right):** Green = healthy high-REP areas. Red/orange = lower REP, potentially stressed or sparse vegetation. Only pixels with NDVI > 0.2 are shown. **Chlorophyll depth map (Fig 15b):** Bright green = higher chlorophyll absorption = more photosynthetically active. Non-vegetation pixels are masked out (shown as white).

Key Findings

- The red-edge position histogram reveals the distribution of plant health across all vegetated campus pixels.
- Areas with high REP values (close to 735–740 nm) correspond to well-maintained, heavily irrigated vegetation such as sports fields or garden beds.
- Lower REP values near 700–710 nm identify potentially stressed vegetation — possibly areas receiving less water or experiencing heat stress from nearby paved surfaces.
- The chlorophyll depth map shows that tree canopies generally have deeper chlorophyll absorption than ground-level grass — trees have more leaf layers contributing to absorption.
- These two metrics together can guide campus groundskeeping decisions — identifying which green areas need more water or fertilisation.

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Section 16 — Spectral Angle Mapper (SAM)

What is this analysis?

Spectral Angle Mapper is a classification and comparison technique that measures the 'angle' between two spectral fingerprints in high-dimensional space. Think of it like comparing the shape of two mountain ranges: even if one mountain range is taller overall (brighter pixel), the shapes can still be the same. SAM ignores brightness differences and only compares spectral shape. A small angle means the pixel closely resembles the reference material. A large angle means it is spectrally very different. Here, SAM was computed against the mean spectrum of each NDVI-defined land-cover class.

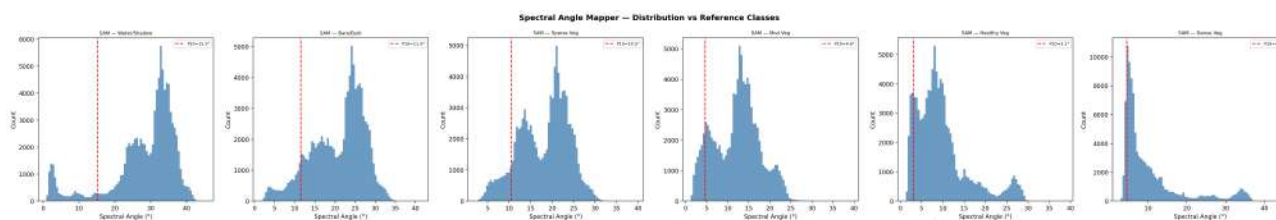


Figure 16: SAM spectral angle distributions for each reference land-cover class. Small angles = close spectral match.

How to read this result

Each panel is a histogram showing, for all sampled pixels, the spectral angle (in degrees) to one reference class spectrum. **A peak at small angles (near 0°)** means many pixels closely match that reference — that material is abundant in the scene. **The red dashed line** marks the 10th percentile — the best-matching 10% of pixels. Narrow histograms = spectrally pure reference. Wide histograms = that reference is spectrally mixed or similar to other materials.

Key Findings

- The vegetation reference classes (moderate and dense vegetation) show narrow SAM distributions with many pixels at small angles — confirming abundant vegetation cover on campus.
- The bare/built-up reference shows a broad distribution, reflecting the spectral diversity of artificial surfaces (different roofing materials, pavement types, concrete ages).
- Water/shadow reference pixels have small angles from a very limited subset of pixels — consistent with the expected scarcity of true water bodies on a campus.
- SAM results are consistent with K-Means and Random Forest classifications, providing independent validation of the spectral groupings.
- SAM's angle-based similarity measure is illumination-independent — useful for comparing pixels in sun vs. shade without misclassifying shadows as different materials.

■ Section 17 — Spatial Texture Analysis

■ What is this analysis?

Texture refers to the spatial pattern of variation in an image — how 'rough' or 'smooth' the pixel values appear at a local scale. A dense forest canopy has high texture because shadows and sunlit leaves create strong local variation. A flat rooftop has low texture because reflectance is uniform. Here, local standard deviation (measured over a 9×9 pixel window) was computed for both the NDVI and the grayscale RGB image. This captures structural complexity that spectral indices alone cannot reveal — helping distinguish tree canopies from flat grass lawns, or corrugated metal roofs from smooth concrete.

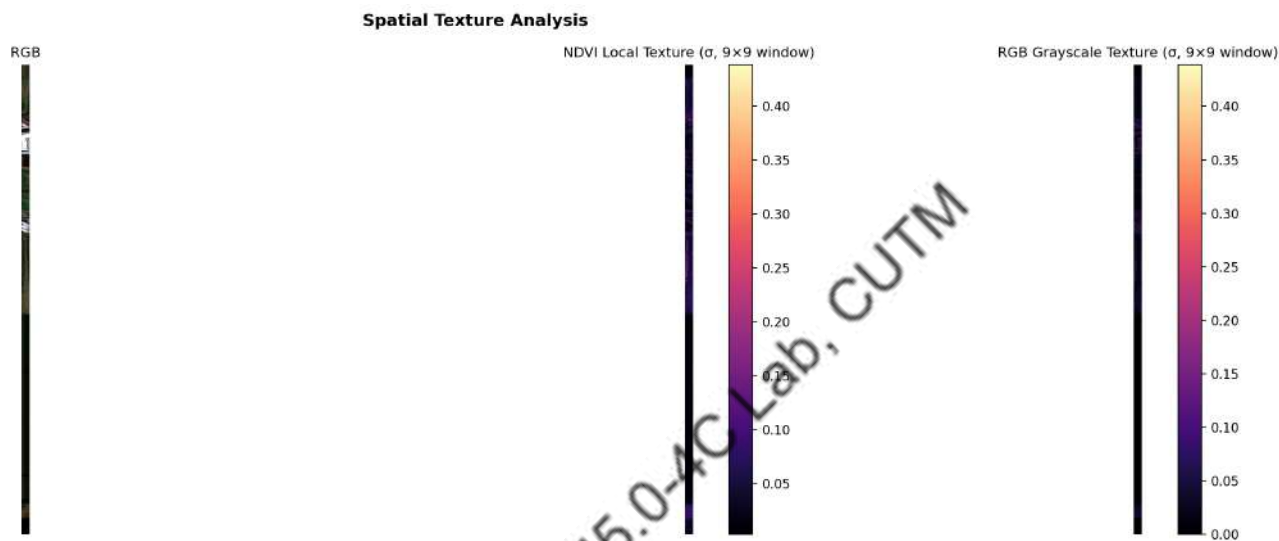


Figure 17: RGB image (left), NDVI local texture map (centre), and RGB grayscale texture map (right). Brighter = more textured.

■ How to read this result

Left panel (RGB): The regular aerial photograph for spatial reference. **Centre panel (NDVI Texture):** Bright areas have high local NDVI variation — typically edges between vegetation and non-vegetation, or within complex tree canopies. Dark areas have uniform NDVI — flat lawns, open soil, or homogeneous rooftops. **Right panel (RGB Texture):** Captures textural contrast from brightness patterns. Building edges, tree canopy borders, and textured surfaces (gravel paths, car parks) appear bright. Flat, uniform surfaces appear dark.

Key Findings

- Tree canopy edges and boundaries between vegetation and built surfaces show the highest texture values — confirming that structural complexity is highest at transition zones.
- Large flat rooftops and parking areas show very low texture — consistent with spectrally and spatially homogeneous artificial surfaces.
- Lawn areas show moderate, uniform texture — lower than tree canopies because individual grass blades are too small to create large-scale variation at this resolution.
- NDVI texture and RGB texture maps show complementary patterns — combining both could improve automated building vs. tree-canopy separation.

- Texture features can be integrated with spectral features in future Random Forest models to further improve classification accuracy.

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■ Section 18 — Summary Dashboard

■ What is this?

The summary dashboard brings together the most important results from all analyses into a single visual overview. It is designed to give a quick, holistic picture of the campus — who is where, how healthy the vegetation is, and where unusual features exist. Think of it as the 'executive summary' of the entire hyperspectral analysis — everything at a glance.

Hyperspectral VNIR Analysis — University Campus Summary Dashboard

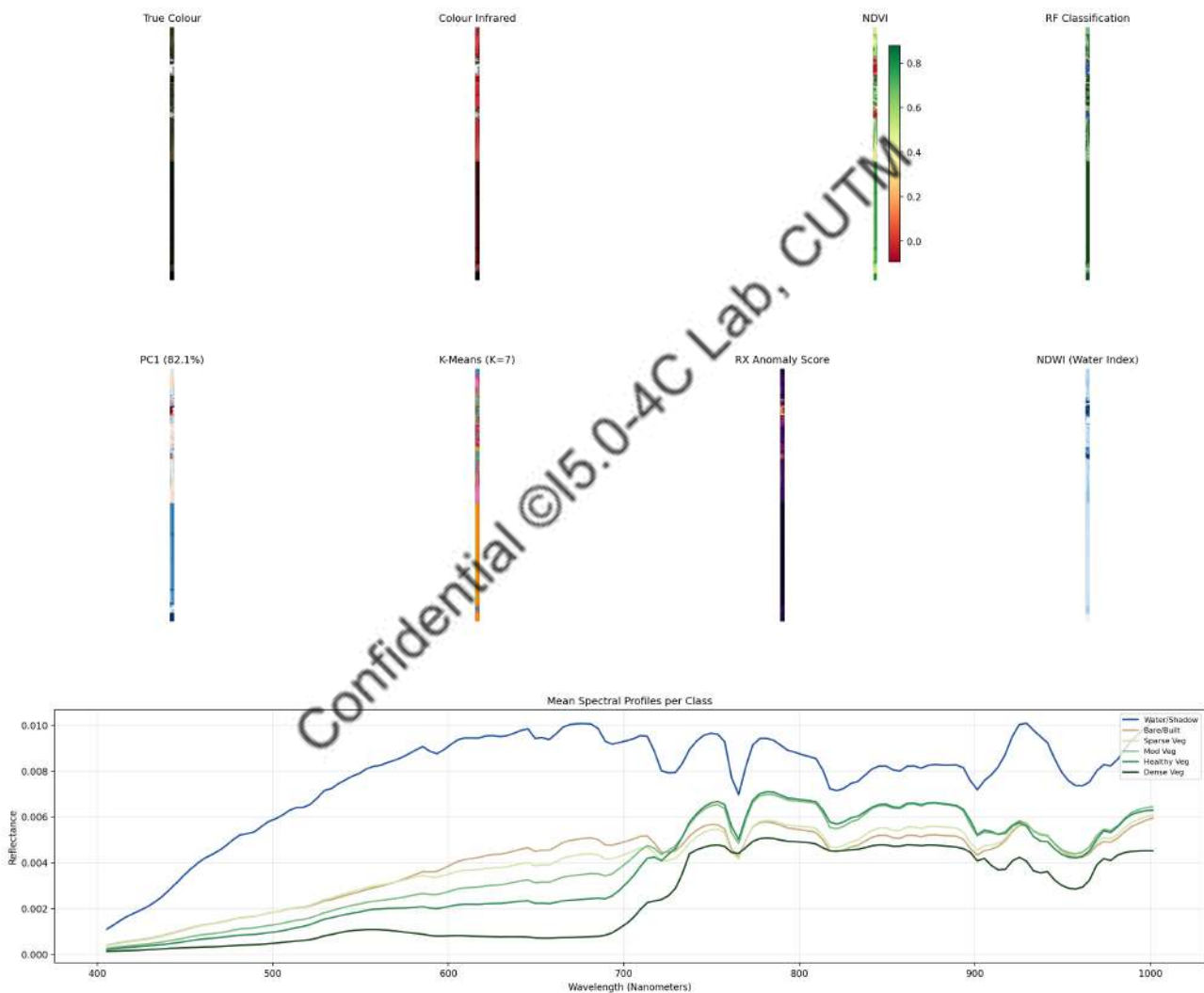


Figure 18: Complete analysis summary dashboard — 8 key outputs and spectral profiles in one view.

■ How to read this dashboard

Top row: True Colour (aerial view), Colour Infrared (vegetation in red), NDVI map, Random Forest Classification. **Middle row:** PC1 pattern, K-Means clusters, RX Anomaly map, NDWI (water index). **Bottom panel:** Spectral signatures of each land-cover class across the full wavelength range — the definitive spectral fingerprints of the campus materials.

Key Findings

- The dashboard confirms agreement across all analysis methods — vegetation, soil, and built areas are consistently identified by independent techniques.
- The spectral profile panel at the bottom is the most scientifically rigorous summary — the distinct shapes confirm well-separated, physically meaningful classes.
- RX anomalies visible in the middle row are spatially specific and do not correspond to any expected land-cover class, warranting further investigation.
- NDWI (water index) in the bottom-right of the middle row reveals the distribution of moisture across the campus, complementing the vegetation maps.
- The full pipeline — from raw hyperspectral cube to this dashboard — demonstrates the power of airborne imaging spectroscopy for urban environmental monitoring.

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■ Conclusions & Recommendations

Overall Findings

■ Campus Vegetation Health

The hyperspectral analysis confirms that the university campus has a mix of healthy and moderately stressed vegetation. Tree canopies show strong near-infrared reflectance and deep red-edge positions, indicating good health. Some lawn areas show lower chlorophyll absorption, which may indicate water stress or nutrient deficiency and could benefit from targeted irrigation or fertilisation.

■ Built Environment Mapping

Rooftops, roads, and paved areas are cleanly separated from vegetated zones across all classification methods. The spectral diversity of building materials is captured in the GMM and K-Means outputs, potentially enabling roof material type mapping with additional analysis.

■ Spectral Anomalies

A small number of spectrally anomalous pixels were detected by the RX detector. These merit follow-up ground investigation to determine if they represent unusual materials, equipment, vehicles, or sensor artefacts. Their spatial clustering suggests a real physical cause.

■ Data Quality

MNF analysis confirms good sensor data quality with clear signal in the first 5+ components. The IGM geocorrection file enables accurate mapping of all results to real-world coordinates, making the outputs directly usable in GIS applications.

■ Glossary of Key Terms

NDVI	Normalised Difference Vegetation Index — a number between -1 and +1 that measures how much live green vegetation is present. Higher = more vegetation.
NIR	Near-Infrared — wavelengths of light just beyond what human eyes can see (~700–1000 nm). Plants reflect NIR very strongly, making it essential for vegetation analysis.
Red Edge	The steep rise in plant reflectance between 680 and 750 nm. Its exact position shifts depending on chlorophyll content — a key plant health indicator.
Endmember	The spectral signature of a pure material (pure grass, pure concrete, pure water). Real pixels are usually mixtures of these.
PCA	Principal Component Analysis — compresses hundreds of bands into a few key 'super-bands' capturing the most important variation.
MNF	Minimum Noise Fraction — like PCA but specifically separates real signal from sensor noise.
t-SNE / UMAP	Non-linear algorithms that visualise high-dimensional spectral data in 2D, grouping similar pixels together.

SAM	Spectral Angle Mapper — measures similarity between spectra based on their shape (angle), not their brightness.
RX Detector	Reed-Xiaoli anomaly detector — finds pixels that are statistically very different from the average scene spectrum.
Continuum Removal	Flattening a spectrum to its baseline so that absorption features (dips) are easier to see and measure.
IGM	Image Geometry Model — a file containing the geographic coordinates (latitude, longitude, elevation) for every pixel in the image.
Reflectance	The fraction of light that a surface bounces back. Ranges from 0 (absorbs all light) to 1 (reflects all light).

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